

Optimizing Supply Chain Management for Small and Medium-Sized Businesses Using AI: A Comparative Analysis of GPT-4.5, Grok 3, Gemini 2.0, and DeepSeek R-1

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Abstract—Small and medium-sized businesses (SMBs) often operate on limited profit margins. They may lack the financial resources to implement expensive Enterprise Resource Planning (ERP) systems or advanced artificial intelligence (AI) tools. However, the advent of generative AI technologies, such as GPT-4.5, Grok 3, Gemini 2.0, and DeepSeek R-1, presents an opportunity for these businesses to leverage AI-driven insights at a more affordable cost. This paper investigates the effectiveness of these AI models in addressing key supply chain management challenges, specifically focusing on demand planning and logistics. The results provide valuable insights into the most suitable AI tools for SMBs seeking to optimize their supply chain operations with minimal investment. This study offers practical guidance for SMBs to improve operational efficiency and enhance their competitive edge in the marketplace by conducting a comparative analysis of these tools.

Index Terms—Artificial Intelligence, Supply Chain, ChatGPT-4.5, Gemini 2.0, DeepSeek R-1, Grok 3, Demand Planning, Logistics, Business, Generative AI, XGBoost, Random Forest Regression.

I. INTRODUCTION

Small and medium-sized businesses (SMBs) across the globe often operate on tight profit margins, with limited financial resources to invest in expensive Enterprise Resource Planning (ERP) systems such as SAP or Oracle. Similarly, sophisticated artificial intelligence (AI) tools like Palantir Foundry and DataRobot, commonly used by large enterprises, are often financially inaccessible to many SMBs. However, the rapid advancement of generative AI technologies, including GPT-4.5, Grok 3, Gemini 2.0, and DeepSeek R-1, offers SMBs an opportunity to leverage AI-driven insights at a reduced cost. Integrating such AI tools into their operations allows SMBs to make more informed decisions, ultimately

improving revenue and reducing operational costs.

A critical area where AI can assist SMBs is in supply chain management. Traditionally, supply chain optimization has relied on complex and costly software solutions, limiting smaller businesses' ability to capitalize fully on data-driven decision-making. However, with the rise of AI, SMBs can now access advanced analytics and automation capabilities without significant financial investments. AI applications in supply chain management encompass various domains, such as demand planning and logistics. By improving these functions, businesses can increase efficiency, reduce waste, optimize inventory levels, and streamline the movement of goods, thereby strengthening overall competitiveness.

This paper aims to evaluate the effectiveness of several AI models—GPT-4.5, Grok 3, Gemini 2.0, and DeepSeek R-1—in addressing key supply chain challenges. Specifically, the study will assess these AI tools by presenting one question from each critical domain of supply chain management: demand planning and logistics. The responses generated by these models will be analyzed to determine which AI tool provides the most accurate and actionable insights. The findings will help identify the most suitable AI algorithms for SMBs seeking to optimize their supply chain operations. This comparative analysis aims to bridge the gap between AI technology and practical business applications, offering valuable guidance for SMBs looking for cost-effective solutions to improve supply chain efficiency while looking at the current capability of these AI tools.

II. LITERATURE REVIEW

Artificial Intelligence (AI) has revolutionized a wide range of industries by enabling machines to perform tasks that require human-like cognitive abilities, such as understanding text, recognizing objects, and making predictions. The increasing availability of computational power and data storage has made AI technologies more accessible, opening up opportunities for substantial economic and social advancements across various sectors. Within this broader field of AI, generative AI, an advanced technology, has gained significant attention, particularly within supply chain management (SCM) [1]. Integrating generative AI into supply chains presents vast potential for improving operational efficiency, enhancing decision-making, and optimizing overall performance. By leveraging generative models, businesses can streamline operations, improve responsiveness, and make more accurate predictions regarding demand, inventory levels, and logistics. Moreover, organizational learning plays a pivotal role in accelerating AI adoption. As businesses become more familiar with AI technologies, they are better equipped to overcome initial challenges and realize the long-term benefits of AI [2]. This increasing familiarity positions companies to harness AI to boost operational efficiency and strengthen their competitive advantage within SCM.

Managing supply chains can be particularly complex for small and medium-sized businesses (SMBs). SMBs often face resource constraints and a lack of specialized expertise, which pressure them to remain competitive. To stay viable, these businesses typically focus on product diversification, supply chain efficiency, and competitive pricing strategies. The advent of generative AI offers SMBs a valuable opportunity to refine these strategies. By leveraging AI technologies, SMBs can enhance decision-making, improve operational efficiency, and respond more swiftly to shifting market demands [3]. This ability to adapt and optimize positions SMBs to improve productivity and strengthen their market standing.

AI applications in supply chain management span many areas, including demand forecasting, inventory optimization, warehouse operations, transportation routing, and logistics management [4]. Demand planning, in particular, is crucial for optimizing business operations. It involves using forecasts to guide strategic decisions, ensuring the smooth movement of raw materials, components, and finished products between suppliers and customers. Effective demand planning enables businesses to evaluate profit potential, make informed decisions regarding capacity, and maintain strong relationships with stakeholders and financial entities [5]. By using robust demand planning methods, businesses can generate more accurate forecasts, leading to improved production volumes, resource allocation, and inventory management decisions. These practices are vital for maximizing profitability and ensuring the efficient use of resources [6]. AI significantly

enhances the analytical capabilities required for precise demand forecasting, helping businesses stay competitive in a fast-paced market environment [7].

The Vehicle Routing Problem (VRP), central to distribution management, presents a significant challenge for many companies involved in goods and people delivery [8]. The VRP aims to optimize vehicle routes and allocations to minimize transportation costs [9]. However, due to the variability in conditions across different settings, the objectives and constraints involved in VRP solutions can differ greatly. Most algorithmic research and software development in this field focus on a limited set of prototype problems, often overlooking the diverse practical contexts businesses face. By introducing greater flexibility into optimization systems, it becomes possible to adapt these solutions to various practical applications [10]. AI technologies have the potential to offer advanced solutions to VRP, enhancing transportation efficiency and helping companies optimize their distribution networks.

Previous research has explored how AI-driven demand forecasting can be integrated into SAP-based supply chain systems, highlighting its ability to improve inventory control through predictive analytics. These studies also address challenges associated with AI implementation, including data integration issues, ensuring model accuracy, and navigating organizational change. They also point to emerging developments in AI-enabled supply chain optimization [11].

Meanwhile, another study concludes that artificial intelligence can significantly boost supply chain efficiency through advanced simulations, predictive tools, and innovative data modeling. It discusses both the opportunities and hurdles of using generative AI in demand forecasting and considers its broader impact on the future of supply chain management. Despite these insights, the literature emphasizes the need for comparative analysis of generative AI models to evaluate their accuracy and performance [12].

III. METHODOLOGY

In this study, various Generative AI tools are presented with one problem from each demand planning and logistics domain.

A. Demand Planning

This study utilizes Kaggle data for demand forecasting in the context of Zara [13]. The dataset is randomly split into a training set (80%) and a test set (20%) [14]. Four Generative AI models—GPT-4.5, Grok 3, Gemini 2.0, and DeepSeek R-1—are evaluated based on their performance in predicting demand. The models' accuracy is assessed using the Mean Absolute Percentage Error for the test dataset (MAPE). Additionally, the recommended machine learning (ML) models for demand forecasting are analyzed

based on MAPE

MAPE (Mean Absolute Percentage Error) is a metric used to measure the accuracy of a forecasting model. It expresses the average absolute difference between predicted and actual values as a percentage of the actual values. A lower MAPE indicates an accurate model [15].

$$MAPE = \frac{1}{n} * \sum_{t=1}^n \left| \frac{A_t - F_t}{A_t} \right| * 100$$

where A_t is actual value, F_t is forecasted value and n represents number of observations.

Prompt used in this case: “Use the most suitable machine learning model to train on the ‘train’ dataset, where ‘sales volume’ is the target variable. Then, generate demand predictions on the ‘test’ dataset and calculate the Mean Absolute Percentage Error (MAPE).”

The dataset provided contains detailed retail sales data, suitable for research in retail analytics, sales forecasting, and consumer behavior analysis. It includes seven key attributes:

- Product Position: Indicates product placement within the store (Front of Store, End-cap, Aisle).
- Promotion: Binary indicator (Yes/No) specifying whether the product was under promotion.
- Seasonal: Denotes if the product sale was seasonal or not.
- Sales Volume: Number of units sold.
- Price: Selling price of the product.
- Terms: Product category (e.g., jackets, sweaters, shoes, t-shirts, jeans).
- Section: Targeted customer segment (MAN or WOMAN).

The dataset comprises numerous records capturing variations in sales volume influenced by factors such as promotional activities, seasonal effects, pricing strategies, and store positioning. This data can facilitate advanced analysis to determine optimal marketing strategies and pricing decisions for maximizing sales performance. It is particularly valuable for developing predictive models using machine learning algorithms to optimize inventory management, promotional effectiveness, and product placement decisions in retail environments.

B. Logistics

This study employs the Belgium-n50-k10 dataset to evaluate AI-based solutions for the vehicle routing problem [16]. The scenario assumes a single truck with a capacity of 125 units, starting at a depot (0,-1) and making multiple trips to fulfill demand. The objective is to minimize the total cost by optimizing travel distance and the number of trips.

Prompt used in this case, “The ‘Customer’ dataset includes customer names, X and Y coordinates, and their demand values. A truck, starting at the depot located at (0, -1), is responsible for fulfilling all customer demands. The truck

has a capacity of 125 units per trip and must return to the depot to refill before making additional deliveries. The cost is \$1 per mile, and the truck must return to the depot after completing all deliveries. Distance is calculated using the Pythagorean Theorem. Determine the minimum total cost (in miles) and report the total number of trips the truck makes.”

The dataset contains customer information suitable for research purposes, particularly in logistics, transportation, or supply chain optimization studies. The dataset includes the following attributes:

- Customer ID: Unique identifier for each customer, incrementing by intervals of 55, starting from 0.
- Coordinates (X, Y): Geographic coordinates representing customer locations, likely latitude (X) and longitude (Y), within a specific geographic region.
- Demand: Numerical values representing the quantity demanded by each customer, ranging from 1 to 33 units.

IV. RESULTS

A. Demand Planning

The comprehensive results of the demand planning analysis, as summarized in Table I, reveal that DeepSeek R-1 outperformed all other models, demonstrating a significant improvement in predictive accuracy. Specifically, when utilizing an XGBoost regressor autonomously as its best choice, DeepSeek R-1 achieved a remarkably low Mean Absolute Percentage Error (MAPE) of 18.54% , which is the best performance among all tested Generative AI tools. In stark contrast, the alternative tools, which choose Random Forest regressors, yielded MAPE values that varied between 23.76% and 25.48%, thereby indicating a substantial performance gap. This pronounced difference highlights the superior predictive power and overall effectiveness of the XGBoost-based DeepSeek R-1 model compared to the Random Forest-based models, further underscoring the advantage of using DeepSeek R-1 for more accurate and reliable demand planning.

TABLE I
DEMAND PLANNING RESULTS

Generative AI Model	ML Model (Autonomously Selected)	MAPE
GPT 4.5	Random Forest Regressor	25.48%
Grok 3	Random Forest Regressor	24.08%
Deepseek R-1	XGBoost Regressor	18.54%
Gemini 2.0	Random Forest Regressor	23.76%

To explain both the machine learning models chosen by generative AIs, XGBoost and Random Forest are both ensemble techniques, but they differ in their model construction strategies.

Random Forest creates many decision trees independently using a method called bagging. It selects random subsets of

data and features for each tree and combines their outputs to improve stability and reduce overfitting [17]. XGBoost, or Extreme Gradient Boosting, builds trees in a sequence. Each new tree aims to fix the errors of the previous ones using gradient descent to minimize loss. It also includes regularization, which helps boost accuracy and efficiency [18].

In general, XGBoost tends to outperform on structured datasets. Meanwhile, Random Forest is more straightforward to implement and tune, making it a good starting point for many problems.

B. Logistics

None of the artificial intelligence (AI) models evaluated in this study successfully solved the vehicle routing problem (VRP), primarily due to the inherent computational complexity associated with this problem. The VRP involves finding the most optimal routes for vehicles to deliver goods to various locations while minimizing costs, a well-known logistics and supply chain management challenge. Despite the advanced capabilities of modern AI tools, the computational demands required to solve VRP instances at scale remain a significant barrier. This highlights the need for continued advancements in AI-powered logistics solutions, especially in the context of small and medium-sized businesses (SMBs), which may face additional resource constraints.

V. CONCLUSION AND DISCUSSION

The inability of current AI models to effectively address the VRP points to the need for further advancements in AI-powered logistics solutions. Future research should focus on developing more scalable and efficient algorithms to tackle the VRP and other logistics challenges. Hybrid AI approaches, which combine multiple AI models or integrate AI with traditional optimization methods, could be an effective solution to improve the scalability and efficiency of logistics operations. Furthermore, AI's ability to solve VRP in real-time, under dynamic and changing conditions, remains a significant area for improvement. Additionally, further studies should explore larger datasets for demand planning and implement more examples using k-fold cross-validation to develop better solutions.

In conclusion, while DeepSeek R-1 demonstrated considerable promise in demand planning, providing SMBs with a tool to enhance forecasting accuracy and optimize supply chain operations, the challenge of solving complex logistics problems, such as the Vehicle Routing Problem, remains unsolved by current AI models. This study suggests that SMBs should consider DeepSeek R-1 for demand planning and inventory management while keeping a close eye on future advancements in AI for logistics. As AI technology evolves, these tools have significant potential to support SMBs in tackling increasingly complex supply chain challenges. Thus, continued investment in AI research and development

is essential to unlocking its full potential for SMBs across various supply chain domains.

REFERENCES

- [1] Anantrasirichai, N., and David Bull. "Artificial Intelligence in the Creative Industries: A Review." *Artificial Intelligence Review*, vol. 55, 2022, pp. 589–656. Springer, <https://doi.org/10.1007/s10462-021-10039-7>.
- [2] Haddud, A. "ChatGPT in Supply Chains: Exploring Potential Applications, Benefits and Challenges." *Journal of Manufacturing Technology Management*, vol. 35, no. 7, 2024, pp. 1293–1312. <https://doi.org/10.1108/JMTM-02-2024-0075>.
- [3] Wamba, S. F., M. M. Queiroz, C. J. C. Jabbour, and C. Shi. "Are Generative AI and ChatGPT Game Changers for 21st-Century Operations and Supply Chain Excellence?" *International Journal of Production Economics*, vol. 265, 2023, p. 109015.
- [4] Mugambe, P. "Financial Innovation for Survival Among Micro-Enterprises with Declining Profit Margins." *Open Journal of Business and Management*, vol. 12, 2024, pp. 3145–3165.
- [5] Toorajipour, R., et al. "Artificial Intelligence in Supply Chain Management: A Systematic Literature Review." *Journal of Business Research*, vol. 122, 2021, pp. 502–517.
- [6] Sheldon, D. H. *World Class Sales and Operations Planning: A Guide to Successful Implementation and Robust Execution*. 1st ed., J. Ross Publishing, 2006.
- [7] Vlckova, V., and M. Patak. "Role of Demand Planning in Business Process Management." *Proceedings of the 6th International Scientific Conference Business and Management*, 2010.
- [8] de Mattos, C. A., et al. "Artificial Intelligence Capabilities for Demand Planning Process." *Logistics*, vol. 8, no. 2, 2024, p. 53. <https://doi.org/10.3390/logistics8020053>.
- [9] Pandiyan, Manikandan, et al. "Using Machine Learning Algorithms to Predict Steam-Assisted Flare Performance." *American Flame Research Committee Conference*, 2023, Denver, CO.
- [10] Vyakarnam, S., G. Jayakumar, and J. A. Cherukara. "Design and Optimization of an Eco-Efficient Vehicle Routing Model for Heterogeneous Fleets and Multi Depot Systems." *Procedia Computer Science*, vol. 230, 2023, pp. 629–640.
- [11] Olasehinde, Tolamise. *The Future of Demand Planning: Integrating Generative AI into Supply Chain Strategies*. 2024.
- [12] Macron, Tolu. *AI-Powered Demand Forecasting and Inventory Management in SAP-Based Supply Chains*. 2025.
- [13] Chinson, S. "Zara Sales." *Kaggle*, <https://www.kaggle.com/datasets/xontoloyo/data-penjualan-zara>.
- [14] Haque, M. A., et al. "Implication of Different Data Split Ratios on the Performance of Models in Price Prediction of Used Vehicles Using Regression Analysis." *Data and Metadata*, vol. 3, 2024, p. 425.
- [15] de Myttenaere, A., B. Golden, B. Le Grand, and F. Rossi. "Mean Absolute Percentage Error for Regression Models." *Neurocomputing*, vol. 192, 2016, pp. 38–48.
- [16] *Timefold Solver 0.8.42: Vehicle Routing*. <https://docs.timefold.ai/timefold-solver/0.8.x/use-cases-and-examples/vehicle-routing/>.
- [17] Chen, Tianqi, and Carlos Guestrin. "XGBoost: A Scalable Tree Boosting System." *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794. <https://doi.org/10.1145/2939672.2939785>.
- [18] Breiman, Leo. "Random Forests." *Machine Learning*, vol. 45, no. 1, 2001, pp. 5–32. <https://doi.org/10.1023/A:1010933404324>.